

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2, BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks: 30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

(192472188)

I M. Madhulika (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Madhulika

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

Q1. Internet Connectivity for a Remote Rural Branch Office

1. Problem Understanding

A multinational company is opening a branch office in a remote rural area where high-speed wired Internet is unavailable. The office needs reliable Internet connectivity for cloud applications, video conferencing, and secure communication with headquarters. Since the budget is limited, the solution must be cost-effective while providing uninterrupted operations and allowing future expansion.

2. Major Constraints

The main constraints are:

- High-speed wired Internet is unavailable.
- Cloud applications require sufficient bandwidth.
- Video conferencing requires stable and low-latency connectivity.
- Secure communication with headquarters is essential.
- The budget for connectivity is limited.
- Business operations should continue with minimum downtime.
- The network should support future growth.

3. Recommended Internet Connection

I recommend using 4G/5G wireless broadband as the primary Internet connection, provided adequate cellular coverage is available.

For uninterrupted operations, a dual-WAN router can be used with:

- Primary connection: 5G/4G broadband.
- Backup connection: Second 4G/5G provider or satellite Internet.

The router can automatically switch to the backup connection if the primary connection fails. This provides better reliability without requiring expensive wired infrastructure.

4. Recommended Networking Devices

Device	Purpose
4G/5G Router	Provides Internet connectivity
Dual-WAN Router	Automatically switches between primary and backup connections
Managed Switch	Connects computers, printers, IP phones, and other devices
Wireless Access Point	Provides Wi-Fi connectivity inside the office
Firewall	Protects the internal network from unauthorized access
VPN Gateway	Creates secure communication with headquarters
UPS	Keeps networking devices operating during short power failures

5. Security Mechanisms

The following security measures should be implemented:

1. **Firewall:** Blocks unauthorized incoming and outgoing traffic.
2. **VPN:** Establishes an encrypted connection between the branch and headquarters.
3. **WPA3:** Protects wireless communication from unauthorized users.
4. **Strong Authentication:** Uses strong passwords and, where possible, multi-factor authentication.
5. **Network Segmentation:** Separates employees, guests, and critical devices using VLANs.
6. **Regular Updates:** Routers, firewalls, and access points should be updated regularly.
7. **Access Control:** Only authorized employees should access company resources.

6. Bandwidth and Reliability

Cloud applications and video conferencing require stable bandwidth. The network should prioritize business-critical traffic using Quality of Service (QoS).

For example:

- Video conferencing → High priority

- Cloud applications → High priority
- Business communication → High priority
- Guest browsing → Lower priority

A backup Internet connection reduces the risk of business interruption.

7. Cost Consideration

A 4G/5G connection is generally more practical than constructing a new wired connection in a remote location. A dual-WAN router provides redundancy without requiring two expensive dedicated connections.

The company can initially deploy the essential devices and add more access points, switches, or higher-capacity connections as the branch grows.

8. Future Scalability

The proposed design can be expanded by:

- Increasing Internet bandwidth.
- Adding additional wireless access points.
- Adding more switches.
- Increasing the number of VPN users.
- Upgrading from 4G to 5G.
- Adding another backup Internet provider.

9. Final Recommendation

A 4G/5G-based dual-WAN network with firewall, VPN, managed switches, secure Wi-Fi, QoS, and backup connectivity is the most suitable solution. It provides a balance between bandwidth, reliability, security, cost, and scalability.

Q2. Wireless Network Design for a University Campus

1. Problem Understanding

The university needs campus-wide wireless Internet access for more than **3,000 students** across classrooms, laboratories, libraries, and hostels. A large number of users may connect simultaneously. Therefore, the network must provide sufficient capacity, minimize interference, maintain security, and operate within a limited budget.

2. Major Constraints

The important constraints are:

- More than 3,000 simultaneous users.
- Large geographical coverage.
- Limited number of access points.
- High user density in classrooms and libraries.
- Wireless interference must be minimized.
- Secure user authentication is required.
- The solution must be cost-effective.
- The network should be scalable.

3. Proposed Wireless Network Design

I recommend deploying **enterprise-grade Wi-Fi 6 access points** connected to managed PoE switches.

The architecture can be represented as:

Internet → Core Router → Firewall → Core Switch → PoE Switches → Wireless Access Points → Students/Staff

A centralized wireless controller or cloud-based management system can be used to configure and monitor all access points.

4. Access Point Placement

Access points should not simply be placed at equal distances. Placement should be based on **user density and building structure**.

Classrooms

Access points should be positioned close to classrooms and lecture halls. High-density classrooms should receive additional capacity.

Laboratories

Laboratories may require dedicated access points because many computers and students can connect simultaneously.

Libraries

Libraries require strong and stable coverage because users may remain connected for long periods and access cloud resources.

Hostels

Access points should be distributed floor-by-floor or zone-by-zone to prevent excessive loading on a single AP.

5. Frequency Band Selection

A combination of **2.4 GHz and 5 GHz**, with support for newer Wi-Fi capabilities where available, should be used.

Band	Advantages	Usage
2.4 GHz	Longer range and better wall penetration	Low-bandwidth/legacy devices
5 GHz	Higher speed and more channels	Main student and staff connectivity

The network should encourage compatible devices to use **5 GHz** to reduce congestion on 2.4 GHz.

Channel planning should be performed to prevent neighboring access points from using overlapping channels unnecessarily.

6. Reducing Signal Interference

Interference can be reduced by:

- Proper AP placement.
- Automatic or planned channel allocation.
- Adjusting transmit power.
- Using 5 GHz for high-density areas.
- Avoiding unnecessary overlap between AP coverage areas.
- Monitoring wireless performance continuously.

7. Authentication and Security

A secure authentication system should be implemented using WPA2/WPA3-Enterprise with 802.1X authentication.

Students and staff can authenticate using university credentials.

Additional security measures include:

- Separate student and staff networks.
- Guest Wi-Fi using a separate VLAN.
- Firewall protection.
- Role-based access control.
- Network monitoring.
- Regular password and credential management.

8. Handling 3,000+ Users

The network should use capacity planning rather than focusing only on coverage.

High-density areas should receive more capacity, while low-density areas can use fewer APs.

Each access point should have a suitable user limit to prevent performance degradation.

Load balancing can distribute users among nearby access points.

9. Cost Optimization

Because the number of access points is limited, they should be installed strategically in areas with the highest user density.

Instead of deploying excessive APs everywhere:

- Prioritize classrooms and laboratories.
- Give high-density areas more capacity.
- Use centralized management.
- Use PoE switches to reduce separate power requirements.
- Monitor utilization and expand only where necessary.

10. Future Scalability

The network should support future expansion by allowing:

- Additional access points.
- Higher-speed switches.
- Increased Internet bandwidth.
- More VLANs.
- New Wi-Fi standards.
- Additional authentication servers.

11. Final Recommendation

The best solution is a centrally managed Wi-Fi 6 campus network using strategically placed access points, 5 GHz as the primary high-capacity band, 2.4 GHz for compatible/legacy devices, WPA2/WPA3-Enterprise authentication, VLAN segmentation, and intelligent channel planning.

This balances coverage, performance, security, interference reduction, cost, and future scalability.

Q3. Secure Network Architecture for a Financial Institution

1. Problem Understanding

A financial institution handles highly sensitive customer and banking information. Therefore, the internal network must provide strong cybersecurity while allowing employees to access authorized customer information and online banking services. The organization must also comply with strict security policies without significantly increasing cost or reducing network performance.

2. Major Constraints

The major constraints are:

- Customer data is highly sensitive.
- Unauthorized access must be prevented.
- Employees need controlled access to business resources.
- Online banking services must remain available.
- Network performance should not be significantly reduced.
- Security policies and compliance requirements must be followed.
- Security solutions should remain within the available budget.
- The design should be scalable.

3. Proposed Secure Network Architecture

A ~~defense-in-depth~~ architecture should be implemented.

Basic architecture:

Internet → Edge Router → Firewall → DMZ → Internal Firewall → Core Switch → VLANs → Servers/Employees

The network should be divided into separate security zones.

4. Network Segmentation

VLANs should be used to separate different types of systems.

VLAN/Zone	Purpose
Employee VLAN	Employee computers
Server VLAN	Internal application servers
Database VLAN	Sensitive customer databases
Guest VLAN	Guest Internet access
Management VLAN	Network device administration
DMZ	Public-facing services

The **database VLAN** should have the strongest access restrictions because it contains sensitive customer information.

5. Firewall Deployment

A next-generation firewall should be placed at the Internet boundary.

It should provide:

- Packet filtering.
- Application control.
- Access control.
- VPN support.
- Intrusion prevention.
- Logging and monitoring.

A DMZ should be used for public-facing services. This prevents Internet users from directly accessing the internal network.

6. Authentication Methods

Strong authentication should be used for employees and administrators.

Recommended methods include:

- Unique user IDs.
- Strong passwords.
- Multi-factor authentication (MFA).
- Role-Based Access Control (RBAC).
- 802.1X authentication for network access.
- Separate administrator accounts.

Employees should receive only the permissions required for their job.

7. Intrusion Detection and Prevention

An IDS/IPS should monitor network traffic for suspicious activity.

The system can identify:

- Unauthorized access attempts.
- Malware traffic.
- Port scanning.
- Suspicious network behavior.
- Repeated failed login attempts.
- Known attack patterns.

The IPS can automatically block or prevent detected malicious traffic.

8. Protecting Sensitive Customer Data

Sensitive databases should not be directly accessible from employee networks.

A controlled flow should be used:

Employee → Application Server → Database Server

Instead of:

Employee → Direct Database Access

This reduces the risk of unauthorized database access.

Data should also be protected using encryption both during transmission and, where appropriate, while stored.

9. Performance Considerations

Security mechanisms should be configured carefully so that they do not become network bottlenecks.

Performance can be maintained through:

- High-performance firewalls.
- Efficient VLAN design.
- Appropriate bandwidth allocation.
- QoS for critical applications.
- Load balancing where required.
- Monitoring network utilization.

10. Compliance Considerations

The institution should maintain:

- Access logs.
- Firewall logs.
- Authentication records.
- Security monitoring.
- Regular vulnerability assessments.
- Backup and recovery procedures.
- Security policies and audits.

These controls help demonstrate compliance with applicable financial-sector and data-protection requirements.

11. Budget Considerations

To control operational costs, the organization can use centralized security management instead of deploying unnecessary security appliances at every location.

Priority should be given to:

1. Firewall.
2. Network segmentation.
3. MFA.
4. IDS/IPS.
5. Access control.
6. Monitoring and logging.

Existing network infrastructure can be upgraded where practical instead of completely replacing it.

12. Future Scalability

The architecture can be expanded by:

- Adding more VLANs.
- Increasing firewall capacity.
- Adding additional servers.
- Expanding monitoring systems.
- Implementing additional authentication servers.
- Increasing network bandwidth.
- Supporting additional branches and remote users through secure VPN access.